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The frontier of cubesat propulsion: MEMs units



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Nomenclature

$[\epsilon_w]$	Allowable strain
$[\sigma_w]$	Allowable stress
α	Divergent angle
β	Convergent angle
$\Delta P_{distributed}$	Distributed pressure losses
ΔP_{valve}	Pressure drop across the valve
δ_{min}	Minimum wall thickness
$\dot{m}_p$	Propellant mass flow rate
$\dot{m}_{real}$	Augmented mass flow rate accounting for viscous losses in the nozzle
$\dot{Q}$	Heat power
$\dot{q}$	Heat power per unit length
$\dot{Q}_{vap}$	Power to vaporize the water
λ	Nozzle 2D losses
μ_{filter}	Flow viscosity at filter
$\Re$	Universal gas constant
ρ_p	Propellant grain density
ρ_w	Wafer density
ρ_{in}	Pressure at nozzle inlet
ρ_{plenum}	Density inside the plenum
$\rho_{up,valve}$	Density at valve inlet
ε	Nozzle expansion ratio
φ_c	Pressure fluctuation coefficient
a	First Vieille's law coefficient
A_b	Burning area
A_c	Combustion chamber area
A_e	Nozzle exit area
A_t	Nozzle throat area
A_{filter}	Filter area
$A_{orifice}$	Area of the injection orifice
A_{valve}	Valve area
c^*	Characteristic velocity
c_D	Nozzle discharge coefficient

c_F	Thrust coefficient
$c_{D, valve}$	Valve discharge coefficient
d	Distance between MSPTs
D_b	Burning diameter
D_e	Exit diameter
D_t	Throat diameter
$D_{feed, line}$	Diameter of the feeding line
D_{filter}	Filter diameter
D_{hc}	Heating chamber diameter
$D_{orifice}$	Orifice diameter
D_{valve}	Valve diameter
E_w	Wafer Young modulus
F	Thrust
F_{ideal}	1D-nozzle ideal thrust
F_{nom}	Nominal thrust
h_f	Final specific enthalpy
h_i	Initial specific enthalpy
H_p	MSPT Plate height
I_{sp}	Gravimetric Specific Impulse
I_{sp}^{id}	Ideal Gravimetric Specific Impulse
k	Specific heat ratio
Kn	Knudsen number
L_c	Convergent length
L_d	Divergent length
L_p	MSPT Plate length
L_{hc}	Length of the heating chamber
L_{liq}	Part of the heating chamber occupied by liquid water
L_{vap}	Part of the heating chamber occupied by superheated vapour
$leak$	Leakage of the valve
Me	Mach number at the exit of the nozzle
m_u	Single MSPT mass
M_{in}	Mach number at nozzle inlet
m_{leak}	Total mass of propellant lost due to leakage

M_{mol}	Molar Mass
$m_{prop,tot}$	Total propellant mass, accounting for leakage
m_{prop}	Propellant mass
n	Second Vieille's law coefficient
n_{MSPT}	Number of MSPT
n_{MSPT}	Number of thrusters
P_1	Heating chamber pressure
P_a	Ambient pressure
P_c	Combustion chamber pressure
P_e	Pressure at nozzle exit
$P_{down,filter}$	Pressure at filter outlet
$P_{f,tank}$	Final pressure in the tank
p_{filter}	Filter porosity
$P_{i,tank}$	Initial pressure in the tank
P_{in}	Pressure at nozzle inlet
P_{plenum}	Plenum pressure
$P_{up,filter}$	Pressure at filter inlet
$P_{up,valve}$	Pressure at valve inlet
$q_{lat,vap}$	Latent heat of vaporization
R	Gas constant (vapor / nitrogen)
r_b	Burning rate
$r_{hole,filter}$	Radius of filter orifices
Re'	Modified Reynolds
Re_t	Reynolds number at throat
T_1	Heating chamber temperature
t_b	Burning time
T_e	Temperature at the exit of the nozzle
T_f	Flame Temperature
T_{filter}	Filter temperature
t_{filter}	Filter thickness
$t_{mission}$	Duration of the mission
$t_{mission}$	Total mission duration
T_{plenum}	Plenum temperature

T_{sat}	Saturation temperature of water
T_{std}	Standard temperature of water
$T_{up,valve}$	Temperature at valve inlet
u_e	Velocity at nozzle exit
u_{in}	Velocity at nozzle inlet
V_u	Volume per thruster unit
$V_{feed,line}$	Volume of the feeding line
V_{hc}	Volume of the heating chamber
V_{nozzle}	Volume of the nozzle
V_{tank}	Volume of the tank
V_{tot}	Total volume of the propulsion unit
w_b	Web length

Acronyms

AOCS Attitude and Orbit Control System. 1

CGG Cold Gas Generators. 25

CGMP Cold Gas Micro-propulsion. i, 19, 22, 24, 25

CGT Cold Gas Thruster. 1, 19, 20, 23, 24, 26

DRIE Deep Reactive Ion Etching. 9

ECM Electro Chemical Machining. 19

LAAS Laboratoire d'Analyse et d'Architecture des Systèmes. 2

MEMS Micro Electro Mechanical System. 1, 2, 8–10, 19

MEOP Maximum Expected Operating Pressure. 13

MSPT MEMS Solid Propellant Thruster. i, ii, 1–3, 5–8, 26

NS Navier Stokes. 2, 7

RPA Rocket Propulsion Analysis software. 4

VLM Vaporizing Liquid Micro-resistojet. 1, 9, 10, 13–18, 26

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1 Introduction

One of the latest development trends in the design of propulsion systems is miniaturisation. Small spacecrafts, such as cubesats, require propulsion systems capable of providing very low thrust values [1]. The challenges of this task arise not only from the scaling of fluid dynamics, but also from Micro Electro Mechanical System (MEMS) manufacturing technologies. The functionality of this paper stands in the presentation of different MEMS propulsion unit designs and places them in a comparative analysis with respect to the research and experiment results. They are designed to be integrated into a 3U cubesat, with a total mass of 4.5 kg, sharing the same final requirements:

Thrust per unit	Total ΔV
5 mN	0.5 m s^{-1}

Table 1: Required performance.

The designs are focused on being integrated in the cubesat's Attitude and Orbit Control System (AOCS). The MEMS technologies explored in the report are the following:

- **MEMS Solid Propellant Thruster (MSPT)**: this technology involves a solid propellant grain, which, once ignited, produces combustion products that are subsequently expanded through a micro-nozzle.
- **Vaporizing Liquid Micro-resistojet (VLM)**: this kind of architecture takes advantage of a liquid propellant which is vaporized inside a heating chamber and then accelerated through a micro-nozzle. The liquid propellant is stored inside a tank, while the heating is performed through electrical resistors.
- **Cold Gas Thruster (CGT)**: in this solution, the propellant is stored either in solid or gaseous form and eventually sublimated. The thrust is generated through a simple expansion of the inert gas inside a convergent-divergent micro-nozzle.

The remainder of the report is organised as follows: in Section 2, the design of the MSPT solution is presented. Section 3 covers the design of the VLM, followed by the CGT design in Section 4. Section 5 compares and discusses the differences between the solutions found, and draws final conclusions.

2 MEMS Solid Propellant Thrusters

2.1 Introduction and literature review

Unlike other technologies, MSPTs have the advantage of higher reliability, simplicity and absence of leakage within the system. However, their one-shot nature and the lack of control after ignition can be challenging. This problem can be addressed by fabricating arrays of MSPT [2], which can be fired asynchronously to achieve fine control. The main architecture of a MSPT is composed of a propellant layer, ignition powder layer, ignition circuit layer, a small cavity layer, and the nozzle layer on top.

According to NASA studies [3], first researches on micro-thrusters using solid propellants started in the late years of the past century. First studies involved the applicability and performance of the micro-nozzle and experiments were carried out to understand their performances related to gas dynamics using Helium and Nitrogen gases. More recent studies at Laboratoire d'Analyse et d'Architecture des Systèmes (LAAS), mainly revolving around studies from Dr. A. Chaalane and Dr. C. Rossi, explore more in detail the design, fabrication and testing of MEMS thrusters in more confined scales of μm [4] [5] [6]. One of the most important components of a solid rocket propulsion system is the igniter. First studies from NASA used laser ignition, thus the propellant grain had to be suitable for such requirement; while nowadays the igniter layer is composed of electronic circuits with the exploitation of resistances to ignite a powder that starts the combustion of the grain. This opens the path for the application of a larger variety of propellants. Finally, latest improvements in the technology involve increasing the number of thrusters per array and new igniter materials feasibility studies are carried out, for example by researchers in La Sapienza, Rome, Italy [7].

The main problems arising with the scaling of a solid thruster are related to the combustion itself. As a matter of fact, it is possible that the combustion is faced with the quenching phenomenon due to the decrease of heat production per unit volume. Additionally, micro thermal and fluid dynamics can highly affect the performance through increased losses in both heat and viscous terms [7]. The issue regarding the viscous effects is a general problem of the micro-nozzles, as investigated by Lee et al. [2]. In fact, microfluidic dynamics brings the nozzle flow to be increasingly rarefied along the nozzle axis, thus the Knudsen number becomes larger and larger, making the application of Navier Stokes (NS) equations less and less realistic and predictions of the expansion process are harder to be made.

2.2 Hypotheses and design choices

In the definition of the MSPT model, the igniter influence is neglected, focusing exclusively on the design of the propellant grain and the nozzle geometries. Thus, the power budget remains zero. The following hypotheses are made:

- **Steady-state operations:** production of constant thrust without considering any transition in the ignition or in the flame extinguishment. This is a limitation of the adopted model, but it is acceptable as a first approach.
- **Adiabatic walls:** no heat exchange through chamber or nozzle walls is considered even though it is not realistic regarding the surroundings of a single thruster. In fact, the experiments show unwanted thruster ignition due to heat flux coming from adjacent chambers [5].
- **No two-phase flow:** the combustion product is assumed to be completely in gaseous phase. This assumption could be quite far away from reality due to the possible presence of unburnt aluminium oxide (Al_2O_3) in the products.
- **Calorically perfect gas:** specific heat constants do not vary along the expansion process.
- **No thermal management:** no direct study of the heat flux is counted, but distance between adjacent thrusters is sized large enough to prevent any need for thermal management.
- **Continuum flow:** the flow is considered to be a continuum. This hypothesis is verified later by considering the Knudsen number in the throat, where the assumption has the highest risk of not being valid.

- **Isentropic expansion:** the expansion process in the nozzle is assumed to be isentropic.

The architecture of a single thruster is assumed to be cylindrical with conical convergent-divergent nozzle. For the design process, some geometric and chamber properties are fixed with commonly implemented values, as seen in Table 2. Finally the nozzle is to be implemented in vacuum, thus null external pressure is assumed.

Property	Value
P_c	1 MPa
A_e/A_t	5
α	10°
β	70°
n_{MSPT}	900

Table 2: Design choices.

One of the most relevant aspects is the nominal chamber pressure, which normally lies in the range 0.1-30 MPa [2] [8]. The operating chamber pressure chosen is 1 MPa. The area ratio is justified through different observations. First of all, various experiments show the area ratios yielding the best performance in MSPTs applications to be between 1 and 5 [3]. These values suggest that small area ratios optimise performance in micro-nozzles for solid propellants, due to the condensation and particle residues occurring at large expansion ratios [9]. Regarding nozzle angles, past experimental results [5] show that high convergence and low divergence angles lead to better performance, thus justifying their selection. The number of MSPTs has also been a design choice, that will be further discussed in Section 2.4.

2.2.1 Material and machining

Several projects considered epoxy resin as the material for the wafer [1] [10], many others opted for silicon [6] [11]. In this case, MACOR® has been chosen because of its advantageous thermal and mechanical characteristics, as studied in [4]. Its main disadvantage is high density. Micro-Electrical Discharge Machining (Micro-EDM) is the manufacturing process selected. High precision Micro-EDM can process ceramics with submicron precision [12], so a tolerance of 1 μm is defined for the manufacturing accuracy of the throat diameter.

Property	Value
ρ_w	2.52 g cm ⁻³
E_w	66.9 GPa

Table 3: MACOR® material properties [13].

It is essential to check if the material is strong enough to withstand the combustion chamber pressure. The minimum wall thickness is computed following the method presented in [1]:

$$\delta_{min} = \frac{\varphi_c P_c D_b}{2.3[\sigma_w] - \varphi_c P_{max}}, \quad [\sigma_w] = E_w[\epsilon_w] \quad (2.1)$$

The value of $[\sigma_w]$ is 66.9 GPa, with $[\epsilon_w]$ set to 0.001. The value of φ_c is set to 1.2. D_b corresponds to the combustion chamber diameter, computed later. This leads to a minimum value of $\delta_{min} = 5.3 \mu\text{m}$, much lower than the distance considered between thrusters in the final array design.

2.2.2 Propellant

The propellant considered is composed by Ammonium Perchlorate (AP) 68%, Aluminium (AL) 18%, and HTPB Binder 14%. The chemical properties of the components can be found in Table 4, with a resulting grain density of $\rho_P = 1762 \text{ kg m}^{-3}$. Cylindrical grain and constant burning area are considered in the design.

Name	Formula	Density [g cm ⁻³]	Standard enthalpy of formation [kJ mol ⁻¹]
Ammonium Perchlorate	NH_4ClO_4	1.95	-295.77
Aluminium	Al	2.7	0
HTPB	$C_{7,075}H_{10,65}O_{0,223}N_{0,063}$	0.92	-58

Table 4: Propellant components properties.

Taking as a reference the study done for the BARIA motor and using Rocket Propulsion Analysis software (RPA) [14], the coefficients for the Vieille's Law and properties of the product gases are collected in Table 5.

Property	Value
a	$0.0212 \text{ mm s}^{-1} \text{ Pa}^{-n}$
n	0.3823
k	1.1931
M_{mol}	$27.9346 \text{ g mol}^{-1}$

Table 5: Burning rate and product gases properties.

2.3 Modelling approach

In this section, the main relations regarding the model for the thruster are presented. The isentropic expansion is modelled and the exit pressure is easily computed through the imposed expansion area ratio, $\varepsilon = A_e/A_t$, and the chamber pressure P_c .

$$\frac{A_t}{A_e} = \left(\frac{k+1}{2} \right)^{\frac{1}{k-1}} \left(\frac{P_e}{P_c} \right)^{\frac{1}{k}} \sqrt{\frac{k+1}{k-1} \left[1 - \left(\frac{P_e}{P_c} \right)^{\frac{k-1}{k}} \right]} \quad (2.2)$$

Furthermore, divergence losses of the nozzle flow are taken into account in the computation of the thrust coefficient. These losses affect only the dynamic part contributing to the thrust, and they are directly dependent on the the angle chosen for the divergent part.

$$c_F = \lambda \sqrt{\frac{2k^2}{k-1} \left(\frac{2}{k+1} \right)^{\frac{k+1}{k-1}} \left[1 - \left(\frac{P_e}{P_c} \right)^{\frac{k-1}{k}} \right]} + \frac{P_e}{P_c} \frac{A_e}{A_t}, \quad \text{with} \quad \lambda = \frac{1 + \cos \alpha}{2} \quad (2.3)$$

$$c^* = \sqrt{\frac{\Re T_c}{M_{mol}}} \left[k \left(\frac{2}{k+1} \right)^{\frac{k+1}{k-1}} \right]^{-\frac{1}{2}} \quad (2.4)$$

Chamber temperature is found by means of an iterative process between MATLAB® and RPA results. A first guess on the combustion chamber temperature is used to compute the characteristic velocity of the chamber. The contraction area ratio, A_b/A_t , is computed through the MATLAB® model, which is taken as an input for the RPA code to iterate, obtaining a new chamber temperature and flow properties. The process is stopped when the convergence between the results is obtained.

In this process, the Reynolds number at the throat, Re_t , is evaluated to account the viscous losses. A modified number, Re' is computed using the curvature ratio outputted from RPA thruster design. The choice of a throat curvature is supported by the fact that sharp angles are avoided to facilitate the manufacturing process [11]. Finally, the main parameters can be obtained through the following equations, always taking into account the losses in thrust.

$$c_D = 1 - \left(\frac{k+1}{2} \right)^{3/4} \left[3.266 - \frac{2.128}{k+1} \right] Re'^{-\frac{1}{2}} + 0.9428 \cdot \frac{(k-1)(k+2)}{(k+1)^{1/2}} Re'^{-1} \quad (2.5)$$

$$I_{sp} = \frac{c^* c_F}{g_0}, \quad \dot{m}_p = \frac{F_{id}}{I_{sp} g_0}, \quad A_t = \frac{F_{id}}{P_c c_F}, \quad A_b = \frac{\dot{m}_p}{\rho_P r_b} \quad (2.6)$$

2.4 Sizing and array design

Once the nominal design is frozen, the sizing of the system can be carried out. Results regarding the operating point of a single thruster are shown in Table 6. Its nozzle geometry is defined from the computed throat, burning and exit areas, and dimensions are summarized in Table 7.

Property	Value
c^*	1338.4 m s^{-1}
c_F	1.64
T_c	2521 K
P_e	31.86 kPa
I_{sp}	223.3 s
$\dot{m}_p$	2.58 mg s^{-1}

Table 6: Nominal operating point.

Property	Value
A_c/A_t	101.68
D_t	$66 \mu\text{m}$
L_c	$110 \mu\text{m}$
L_d	$234 \mu\text{m}$

Table 7: Nozzle geometry.

Considering nominal operating conditions, the total mass needed for delivering the total ΔV of 0.5 m s^{-1} is computed through Tsiolkovsky equation, resulting in 1.169 g. The final solid rocket propulsive system will consist of an array divided into four plates of MSPTs that are fired independently, in order to achieve fine attitude control (Figure 3). These plates are normally set in a square disposition. Each of them will consist of a 15×15 disposition of MSPTs, with a separation between them of $d = 1 \text{ mm}$, summing up to 900 MSPTs. This separation is more than enough to avoid undesirable ignition of neighbouring thrusters, being MACOR® ceramic the material selected [4].

From the total number of thrusters, the individual propellant mass and chamber volume can be extracted, enabling the computation of the web length, w_b , and the burning time, t_b .

$$w_b = \frac{V_u}{A_b}, \quad t_b = \frac{w_b}{r_b} \quad (2.7)$$

Finally, this allows to define the height of the thruster, H_p , that accounts for the web length, igniter and cavity layers, convergent and divergent lengths. Inspired by the design proposed by researchers from La Sapienza [7], each plate will be able to freely rotate around an axis through a mechanical system, to allow for thrust direction change. Thus, the dimensions of an individual MSPT plate are computed and shown in Table 9. The total length of the array will be approximately doubled, with a slight margin, resulting in a square of about $60 \times 60 \text{ mm}$, which can be effectively integrated in a cubesat. The total volume needed for the propulsive system is around 11.2 mL. This volume does not account for mechanical movement of the array units.

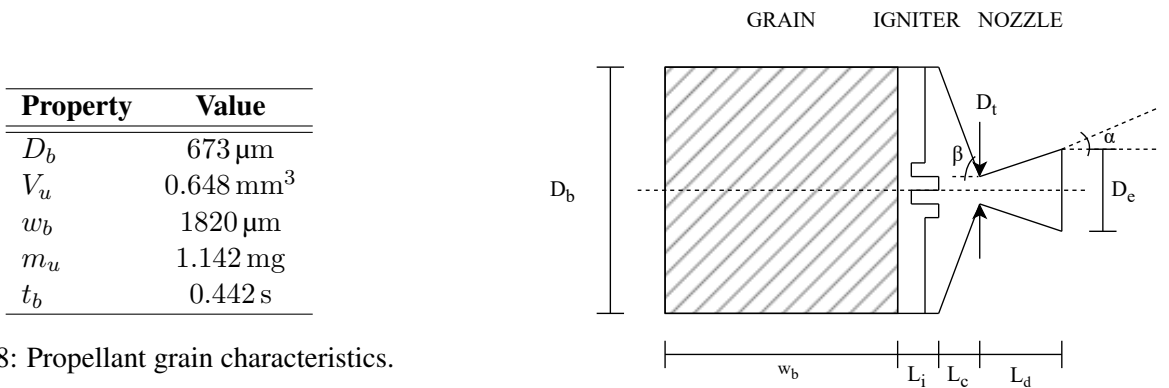


Table 8: Propellant grain characteristics.

Figure 1: MSPT schematic.

Property	Value
L_p	28 mm
H_p	3.2 mm
d	1 mm

Table 9: MSPT plate dimensions.

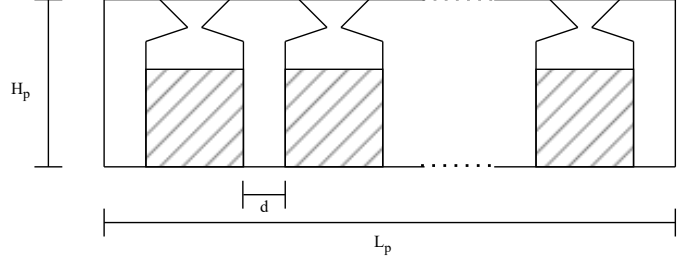


Figure 2: MSPT plate schematic.

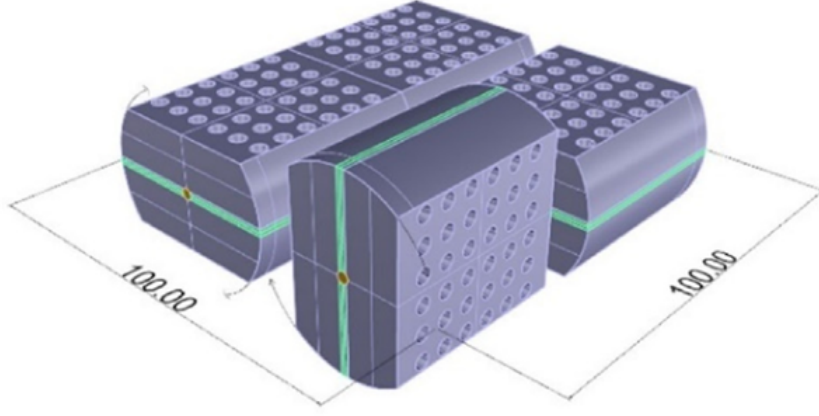


Figure 3: Thruster cluster configuration example [7].

2.5 Results and discussion

The design carried out can now be compared to what is produced through experiments and more advanced mathematical models. Implementations with similar thrust performances use the same propellant and deliver even a slightly larger thrust. Values up to 14 mN can be achieved with slightly larger geometries [11]. In the proposed design, the thrust is mainly reduced by viscous loss, with respect to 2D losses, which are less relevant due to the small angle chosen for the divergent nozzle. Indeed, a significant $c_D = 0.87$ is related to the low Reynolds number in the throat, $Re = 451$.

The burning time per thruster depends not only on the burning rate but also on the volume of the grain. The latter is fixed once the total propellant mass and the number of thrusters is decided. Such decision, aligned with the current technology limitations, allows to have a burning time similar to what can be seen in various experiments [1] [2] [5].

Regarding the specific impulse, the ideal value can be computed by its analytical expression [7]:

$$I_{sp}^{id} = \frac{1}{g_0} \cdot \sqrt{\frac{2k\mathfrak{R}}{k-1} \frac{T_f}{M_{mol}} \left[1 - \left(\frac{P_e}{P_c} \right)^{\frac{k-1}{k}} \right]} \quad (2.8)$$

With a flame temperature of $T_f = 3440$ K, an ideal value of $I_{sp}^{id} = 236.7$ s is yielded, against the actual one $I_{sp} = 223.3$ s. The combustion chamber temperature reached is $T_c = 2521$ K, which lies in the range expected for the assigned propellant [11]. The decrease in performance accounted for by the losses and the lower temperature reached in the combustion chamber, are now clearly appreciable and place the propellant within reasonable range of expected specific impulse [15].

With respect to the number of MSPTs, it was fixed to 900 for two main reasons:

- Having a reasonable ratio between web length and diameter. The ratio achieved is $w_b/D_b \approx 2.7$, that is consistent with values from previous research [6].
- Having a squared array, aligned with the squared ignition circuit proposed in [1]. Thus, a 30x30 disposition of MSPT is defined.

A higher number of MSPT could be considered, but this might introduce problems in terms of the total array size. As discussed previously, distance between thrusters needs to be sufficiently large to avoid undesired ignition of neighbouring thrusters. The proposed design is already 60x60 mm, but a larger number of thrusters might increase these dimensions to a point that the system might not be suitable for a cubesat. Taking advantage of the proposed mechanical system, thrusters could be placed against each other on the plate (Figure 3). Once all the motors of a single plate have been consumed, the plate can be flipped to use the other ones [7].

Finally, the validity of the adopted model is evaluated through the Knudsen number in the throat: $Kn = 0.013$. This number shall be smaller than 0.01 in order for the continuum hypothesis to be acceptable [16]. Thus, implementing slip models to NS might be needed for further analysis.

2.6 Monte Carlo analysis

The performance of the proposed design can be affected by the uncertainties due to the fabrication process. The most impactful among the uncertainties are represented by the Vieille's law parameters and throat area. As presented before, standard deviation values for the parameter a and n are extracted from the experimental data provided by BARIA engine characterisation, whereas the uncertainty on the throat area is introduced by the machining technology. The adopted values are reported in Table 10.

Parameter	Mean value	Standard deviation	Units
a	0.0212	0.00091	$\text{mm s}^{-1} \text{Pa}^{-n}$
n	0.3823	0.00282	-
D_t	66.8	1	μm

Table 10: Monte Carlo inputs for MSPT.

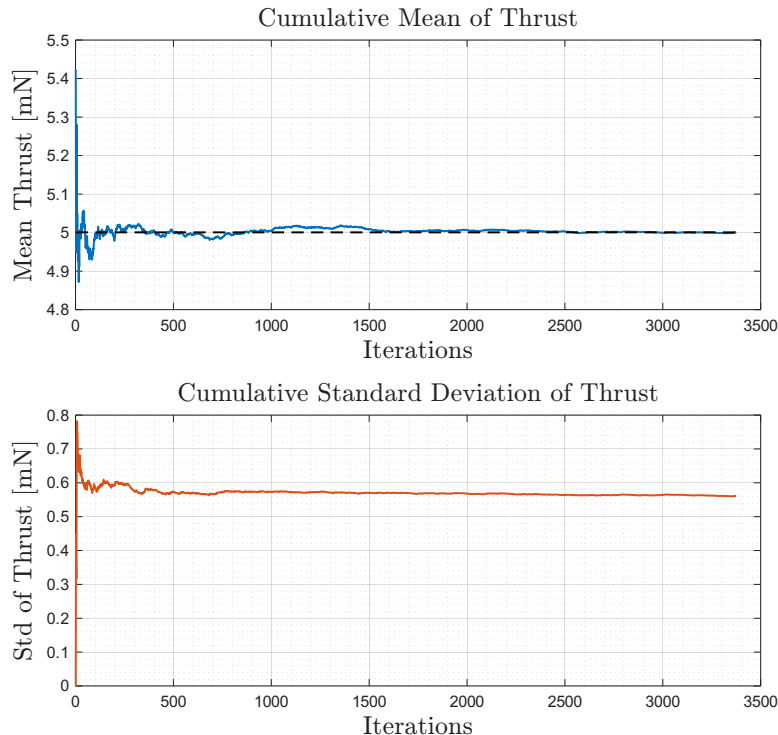


Figure 4: Thrust mean value and standard deviation for MSPT.

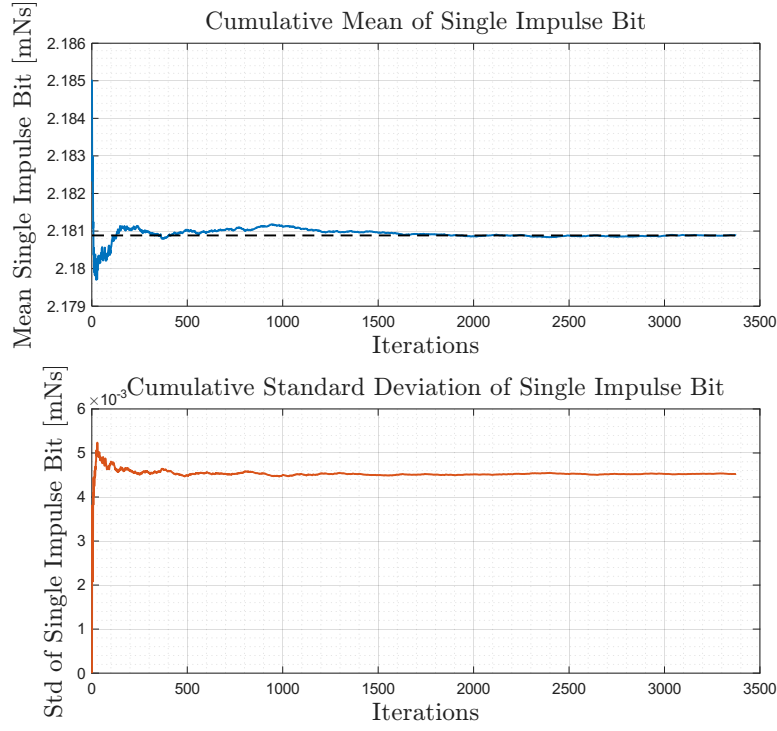


Figure 5: Single impulse bit mean and standard deviation for MSPT.

These deviations from nominal design are exploited to run a Monte Carlo campaign and the following results are found in Figures 4 and 5. The real thrust, accounting for both 2D and viscous losses can be seen to converge to a cumulative mean value around 5 mN with a standard deviation of 0.6 mN. This means that, depending on the building precision and the propellant grain, the thrusters of an array could produce thrust values with relatively large uncertainties.

2.7 Future developments

- **Aluminium-free propellant:** following ESA's Zero Debris Mitigation Guidelines [17], aluminium-free propellant shall be pursued to avoid expulsion of small solid particles as debris, although flame temperature and performance will be decreased. As presented already, powder could be an even larger danger for MEMS due to thruster clogging problems, so it represents another motive to avoid its usage or to research for an alternative solution which could be reasonably scaled also into typical sized propellant grains. In the case under study, the performance of AP-HTPB propellant without aluminium can be augmented by increasing AP content, although a decrease of the mechanical properties is expected and casting technology might be a limitation [18]. Larger expansion ratios might be another option.
- **Igniters circuit design:** although the igniter design was not considered in the analysis, it is one of the most important aspects of MSPTs. Further studies should consider both the computation of the power budget needed and the design of an adequate circuit that allows independent firing of MSPTs in an array.
- **Transient analysis:** the results achieved in this report are poorly represented in experimental data to what concerns the real thrust profile in time [7]. Follow up studies should take into account the transients in the combustion process, aligned to what happens in a real scenario, as seen in experimental results of similar systems [11].
- **Larger scale MSPT arrays:** the feasibility of the integration of a higher number of thruster should be studied. Future missions will require this to achieve finer attitude control, or long term orbit keeping. Feasibility in terms of manufacturing and operation, with a strong focus in the heat exchange between neighbouring thrusters, could be further analysed.

3 Vaporizing Liquid Micro-resistojet

3.1 Introduction and literature review

Among the different MEMS-based micro-thrusters, Vaporizing Liquid Micro-resistojet (VLM) solution can be found. This kind of technology emerged at the end of the 1990s when several studies, particularly from NASA [19] [20], investigated different concepts and experiments aimed at studying the feasibility of this architecture. As explained in Section 1, the working principle of this type of micro-propulsion consists of injecting liquid propellant into a chamber where it is heated and vaporized. Subsequently, the superheated vapor is expanded through a micro-nozzle, creating thrust. Different design solutions that implement this idea are nowadays available. Fontanarosa et al. [21] investigated the most relevant configurations developed during the last thirty years and Zandvliet [22] provided an overview on the cubesats that actually exploit this kind of solution. Figure 6 shows some significant designs and concepts.

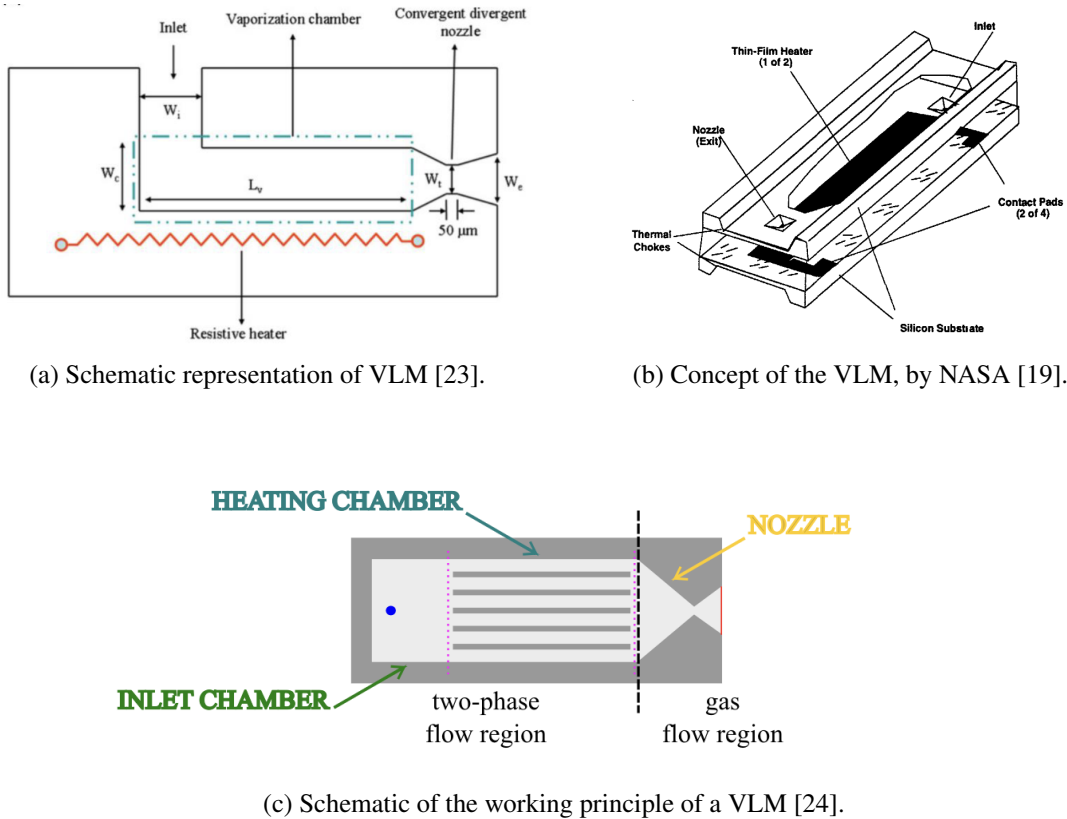


Figure 6: Several conceptual and fabricated designs of VLMs.

The shape of this kind of systems is very different with respect to the one associated to the classic axisymmetric case. This is due to the fact that reduced dimension requirements, related with the compact size of cubesats, have to be met. In this type of technology, the typical lengths range from hundreds of micrometers to a few tens of millimeters. As a result, the manufacturing techniques are extremely critical, as extensively discussed in several papers by van Wees [25], Silva [26] [27], and other researchers from Delft University, which is among the most active institutions in the field of design and performance analysis for this type of technology. The main techniques adopted are Deep Reactive Ion Etching (DRIE) and wafer bonding.

A key aspect of VLMs is the liquid propellant used. In most cases, water is employed due to its ability to provide a high ΔV per unit volume compared to other green propellants. Other possible choices, such as R134a and ammonia, can be adopted, and further options have been investigated by Guerrieri et al. [28] in terms of performances.

A drawback associated to this type of technology is the need for heat to induce the vaporization of the liquid propellant. The required power usually ranges from 2 W to 20 W and is transferred through resistors, typically

made of molybdenum, supplied with current. The possible configurations that heating channels can have, as shown by Silva et al. [26], are various and depend on the specific design used.

Finally, it is noteworthy to say that this type of solution shares the same issues in the modelling of the nozzle flow, as those previously mentioned in Section 2.1, which are common to all MEMS technologies.

Given the design requirements outlined in Table 1, the following analysis will be divided into two parts. In the first part, a parametric analysis will be carried out, varying the input power and the temperatures reached in the heating chamber. This will allow the identification of all the possible designs that meet the specified requirements. Among the feasible configurations, one will then be selected based on a trade-off. In the second part, starting from the previously selected configuration, the sizing of the heating chamber and the feeding system will be performed. Finally, the overall system architecture will be presented.

3.2 Modelling Approach

The sizing and the performance evaluation of the VLM under study can be treated in first approximation using the same equations as in normally sized systems. However, it is of paramount importance to note that the assumptions generally made in the performance analysis of a propulsion system may not be applicable to micro-propulsion systems. Among the previously mentioned assumptions, the most important ones are listed below:

- **Friction forces can be neglected:** viscous losses are neglected in the heating chamber for simplicity, whereas in the nozzle they are negligible as will be discussed later.
- **Continuum flow:** Section 2.2.
- **Calorically perfect gas:** Section 2.2.
- **No heat dispersion through the walls:** the heat is directly transferred to the fluid, without dissipation in the walls.
- **Steady-state operations:** production of constant thrust without considering any transient in the opening or closing of the valves. This is a limitation of the adopted model, but it is acceptable as a first approach.
- **Isentropic expansion:** Section 2.2.

Since the architecture exploits the heat generated by a heater to vaporize water, the first step consists of an energy balance as follows:

$$\dot{Q} = \dot{m}_p (h_f - h_i) \quad (3.1)$$

In the expression above, h_i is the value of the specific enthalpy of water at 25 °C and 3 bar, since this is the chosen operative condition inside the heating chamber, consistent with [25]. h_f is the value of the specific enthalpy of superheated vapor at the end of the heating process. The pressure in the chamber is kept constant by the presence of the plenum, whose main function is to avoid pressure fluctuations. In order to study what happens in terms of sizing and performances in different conditions, the analysis is performed using the final temperature as a free parameter. The corresponding value of h_f is then computed by linearly interpolating the values taken from superheated vapour tables [29]. For the analysis, $\dot{Q}$ has also been considered as a varying parameter in the range between 2 W and 20 W, which turns out to be reasonable considering the state of the art of this technology [22].

By inverting the Eq (3.1), $\dot{m}_p$ can be calculated as function of both the free parameters. After determining the mass flow rate, the next formula can be rearranged to get the throat area A_t , in fact the pressure in the chamber P_1 is given as input and the temperature in the chamber T_1 is known from the previous step.

$$\dot{m}_p = A_t \cdot P_1 \cdot \sqrt{\frac{\left(\frac{2}{k+1}\right)^{\frac{k+1}{k-1}}}{k \cdot R \cdot T_1}} \quad (3.2)$$

The equation above requires the knowledge of the specific heat ratio $k = 1.3$, assumed constant, and of the vapor constant $R = 461.5 \text{ J kg}^{-1} \text{ K}^{-1}$. A further step in the analysis of the motor involves the solution of the following system of 5 equations and 5 unknowns, which must be solved numerically for each possible configuration of the free parameters $\dot{Q}$ and T_f . The system is based on the equations of isentropic expansion, assuming steam as a perfect gas, on the thrust equation and on the relationship between velocity and Mach number. The unknowns that must be computed are the vapor properties at the exit section of the conical nozzle, T_e , P_e , M_e , v_e and the nozzle exit area A_e .

$$\begin{cases} \frac{A_e}{A_t} = \left(\frac{k+1}{2}\right)^{-\frac{k+1}{2(k-1)}} \cdot \left(1 + \frac{k-1}{2} M_e^2\right)^{\frac{k+1}{2(k-1)}} \cdot M_e^{-1} \\ T_e = T_1 \left(1 + \frac{k-1}{2} M_e^2\right)^{-1} \\ P_e = P_1 \left(1 + \frac{k-1}{2} M_e^2\right)^{-\frac{k}{k-1}} \\ v_e = M_e \sqrt{k \cdot R \cdot T_1} \\ F_{nom} = \lambda \dot{m}_p v_e + A_e (P_e - P_a) \end{cases} \quad (3.3)$$

In the system of equations above, the thrust provided must be 5 mN, being the solutions all the possible configurations of the nozzle, in terms of A_t and A_e , which are able to fulfill this constraint. It is worth noting that P_a is assumed to be negligible since the nozzle is expanding in vacuum. Particularly relevant is the presence of the term λ , which accounts for 2D losses in the nozzle. Although in real applications the typical nozzle geometry for micro-propulsion systems is the linear micro-nozzle [16], which is easier to realize from a technological viewpoint, in this treatment the axisymmetric micro-nozzle with circular cross area is employed. This approach ensures both better performances and a simpler treatment [16]. Considered this, the 2D losses can be modelled by using the equation Eq (2.3).

The gravimetric specific impulse and the thrust coefficient can then be computed in the following way:

$$I_{sp} = \frac{F}{\dot{m}_p \cdot g_0}, \quad c_F = \frac{F}{P_1 \cdot A_t}, \quad (3.4)$$

where F is either the real thrust provided by the engine, so to obtain the real specific impulse and thrust coefficient, or the ideal force to get the ideal ones.

It is noteworthy to say that the discharge coefficient of the nozzle, c_D , is considered equal to 1 when dealing with vapor, as stated in [16], which is the reason why the friction losses were neglected. Once the performance parameters are known, a specific configuration is chosen through a trade-off, as it will be better explained in the next section.

The second part of the modelling deals with the sizing of both the heating chamber and the tank. From the computed data, it is immediate to calculate the cross area of the chamber:

$$A_c = \frac{A_t}{M_c} \left[\left(\frac{2}{k+1}\right) \left(1 + \frac{k-1}{2} M_c^2\right) \right]^{\frac{k+1}{2(k-1)}} \quad (3.5)$$

The Mach number in the chamber, M_c , is computed through the data given in [30] and is equal to 0.1. At this point, the knowledge of the chamber, throat and exit areas allows the computation of the nozzle length, once the angles of the convergent and divergent parts are fixed as $\alpha = 30^\circ$ and $\beta = 45^\circ$ [16].

For what regards the sizing of the heating chamber, its total length was assumed as $L_{hc} = 0.001$ m. This length is retrieved by a visual analysis and comparison of images taken from [26]. The volume was therefore recovered as follows:

$$V_{hc} = A_c L_{hc} \quad (3.6)$$

At this point, to check the validity of the results, the residence time of the flow in the heating chamber can be computed. To perform this calculation, it is important to know the value of density, but since in the chamber liquid water is vaporized, the latter is subject to a huge change, since it passes from 950 kg m^{-3} to about 1.3 kg m^{-3} . To use a reasonable value for the density, the total power available was divided by the total length of the heating chamber to get the power available per unit length. Afterwards, the power to vaporize the water was computed to retrieve the percentage of the overall chamber occupied by the liquid. The remaining portion of the chamber was as a consequence occupied by the steam. This process is represented by the following equations:

$$\dot{q} = \frac{\dot{Q}}{L_{hc}} \quad (3.7)$$

$$\dot{Q}_{\text{vap}} = \dot{m}_p [c_p (T_{\text{sat}} - T_{\text{std}}) + q_{\text{lat,vap}}] \quad (3.8)$$

$$L_{\text{liq}} = \frac{\dot{Q}_{\text{vap}}}{\dot{q}} \quad (3.9)$$

$$L_{\text{vap}} = L_{hc} - L_{\text{liq}} \quad (3.10)$$

The residence time is then computed to check the validity of results:

$$t_{\text{res}} = \frac{V_{hc} \cdot \rho_{\text{med}}}{\dot{m}_p} \quad (3.11)$$

The last part of the model deals with the sizing of the tank. The mass of propellant needed for the mission can be computed knowing the requirement on the $\Delta V = 0.5 \text{ m s}^{-1}$. Given a wet mass of 4.5 kg, the dry mass can be obtained by inverting the Tsiolkovsky equation, and therefore the mass of water as the difference between them. Computing the volume flow rate by dividing the mass flow rate by the density of the liquid, it is possible to move upstream and derive the pressure of water before the injection in the chamber across the orifice. Before entering the following equation, the area of the orifice is set as 15% with respect to the area of the heating chamber and the discharge coefficient is computed by means of the Reynolds number. To properly quantify the Reynolds number, the graph found in [31] is interpolated through a spline in MATLAB®.

$$P_{\text{before.inj}} = P_1 + \frac{\rho L}{2} \left(\frac{\dot{m}_{\text{vol}}}{C_d A_{\text{orifice}}} \right)^2 \quad (3.12)$$

The distributed losses are now accounted for by considering the length of the pipe between the injector and the tank valve equal to $2.5 D_c$ and its diameter equal to that of the heating chamber, as shown in one of the possible configurations in [23]. Once the Reynolds number in the pipe is computed, since the flow is laminar, the following equation can be used and provides an estimate of the losses:

$$\Delta P_{\text{distributed}} = \frac{64}{Re} \frac{L_{\text{pipe}}}{D_c} \frac{\rho}{2} \left(\frac{\dot{m}_{\text{vol}}}{A_c} \right)^2 \quad (3.13)$$

For the analysis of the tank, it is important to highlight that a typical pressurizing methodology involves a blowdown system based on Nitrogen (N_2) [22]. This must satisfy the constraint that at the end of the expansion,

the pressure of the gas must be over $\Delta P_{distributed} + P_{before.inj}$. The previously mentioned expansion can be modelled with an adiabatic polytropic equation, applied to the gas between the beginning and the end of the expansion. In addition, the following relation between volumes holds:

$$V_{gas.end} = V_{gas.start} + V_{water} \quad (3.14)$$

These constraints can be translated into a sizing equation in the subsequent way:

$$V_{gas.start} = \frac{V_{water}}{\left[\left(\frac{P_{start}}{P_{end}} \right)^{\frac{1}{\bar{k}_{N_2}}} - 1 \right]} \quad (3.15)$$

A P_{start} of 6 bar is used, as a reasonable Maximum Expected Operating Pressure (MEOP) from literature.

The parametrized analysis developed so far gives all the possible configurations which satisfy the requirement on the thrust. The figure below represents a typical system scheme for a VLM propulsion system whose theoretical modelling has been discussed so far. The relative proportions between the different lengths vary depending on the chosen configuration, as will be further discussed in Section 3.3.

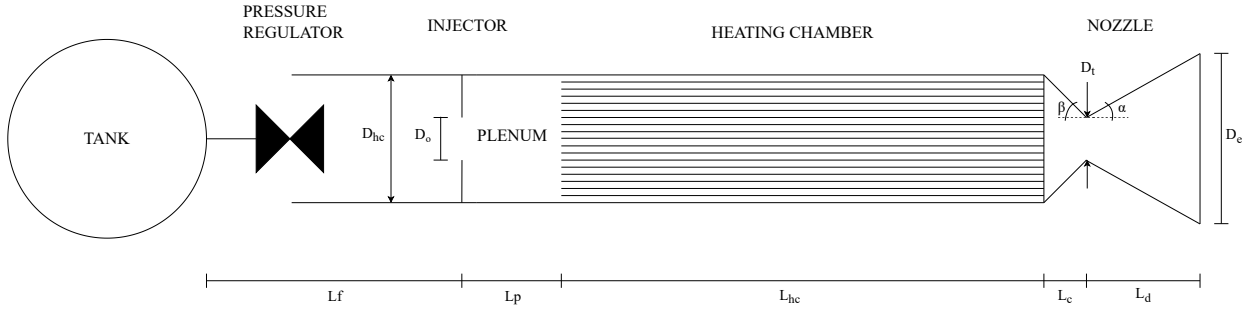
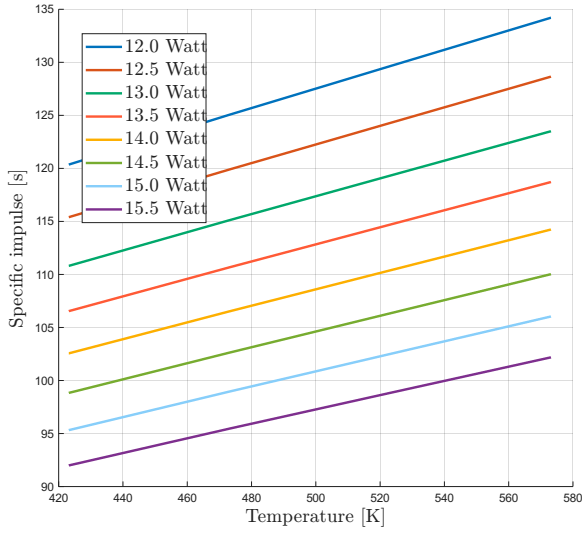


Figure 7: Designed VLM configuration.

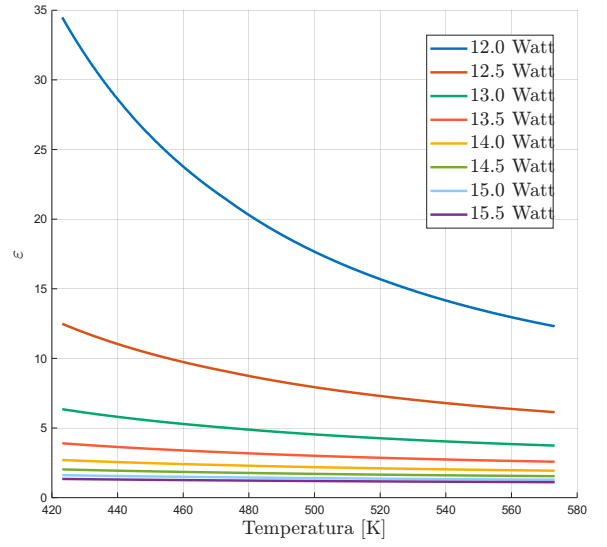
3.3 Results, design choices and performance analysis

Several information and quantities can be retrieved from the initial parametric analysis, but the most significant ones, are shown in the following charts. These will be useful, as previously mentioned, to establish the main design aspects and thus enable the sizing of the entire system. Looking at the charts in Figure 8, a first remark can be made. The parametrization ranges from 12 W to 15.5 W, although in the previous section a broader power range, from 2 W to 20 W, was mentioned. This is due to the fact that, for input powers greater than 15.5 W, the set of Equations (3.3) does not yield a valid solution, as the thrust generated would always exceed the required 5 mN. Conversely, for powers below 12 W, it is not possible to reach a minimum thrust of 5 mN, and thus the system of Equations (3.3) has, again, no solution. Given this, only the meaningful cases corresponding to input powers in the range between 12 W and 15.5 W have been considered and displayed.

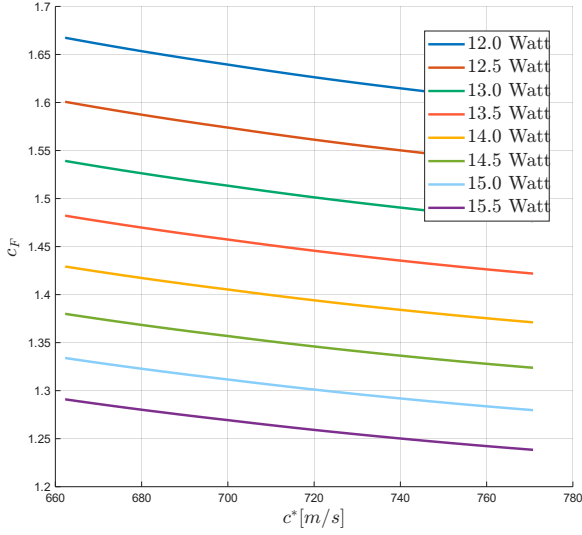
The performances, although very important in the design of propulsion systems, are not always the main criterion of selection. From the previously presented charts, various considerations can be made. First of all, it can be observed that the specific impulse in the case in analysis is between 92 s and 140 s, which is consistent with values from literature [21] [26], given water propellant. It is noteworthy to say that specific impulse decreases as power increases. This would suggest choosing a configuration that exploits the lowest available power.



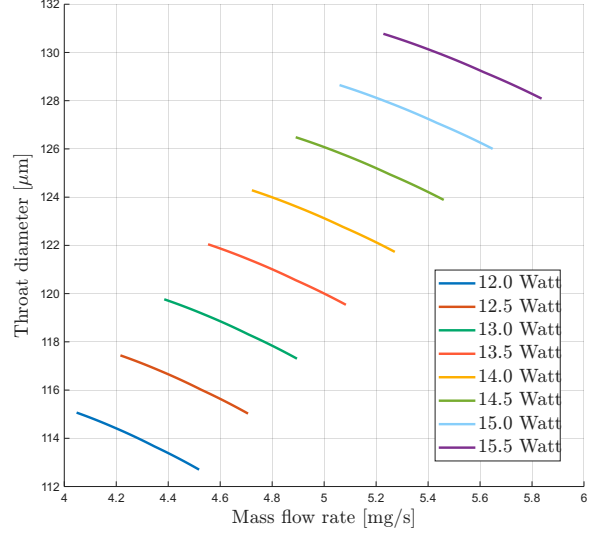
(a) Specific impulse as a function of chamber temperature for different power levels.



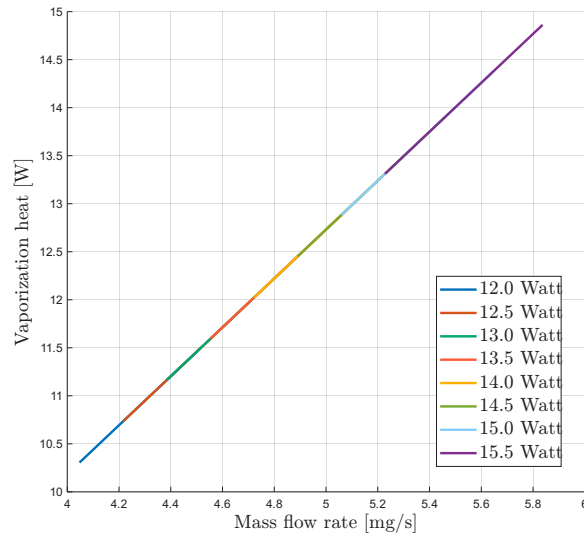
(b) Nozzle expansion ratio (ϵ) as a function of chamber temperature for different power levels.



(c) Thrust coefficient (c_F) as a function of c^* for different power levels.



(d) Required throat diameter as a function of mass flow rate for different power levels.



(e) Required vaporization heat as a function of mass flow rate for different power levels.

Figure 8: Parametric trends of key VLM performances as functions of temperature and input power.

On the other hand, another aspect shall be analysed. From Figure 8b, it is possible to see the possible expansion ratios that, given a specific power and temperature, meet the requirements. These values range from few units to several tens of units, showing a high variability in the potentially adoptable nozzle geometries.

The chart in Figure 8c aims to compare the thrust coefficient and the characteristic velocity for different amounts of heating power. It is peculiar to point out how the characteristic velocity has been computed, even though the propulsion system under study does not exploit any combustion mechanism. This choice is based on the fact that this parameter can still give some insight into the efficiency of the heat transfer mechanisms in the chamber. Moreover, this quantity is put along the x axis since it behaves linearly with temperature.

The amount of power used in the VLM, given the 5 mN thrust requirement, also influences the involved mass flow rate. Figure 8e shows how the latter is related to the heat required for water vaporization. The plot exhibits a linear trend, which is expected due to the structure of Eq (3.1). Conversely, Figure 8d shows that increasing power and mass flow rate demand a larger nozzle throat area. In order to define the main design aspects of the VLM, three representative cases were selected and summarized in Table 11.

Power [W]	I_{sp} [s]	ϵ	$\dot{m}_p$ [mg s ⁻¹]	T [K]
12.0	114.1	28.54	4.46	440.25
12.5	109.5	11.00	4.64	440.25
15.5	88.3	1.30	5.76	440.25

Table 11: Parameters as a function of power.

When dealing with low power levels, even though high specific impulses can be achieved, extremely high expansion ratios are required, which are clearly unfeasible from a technological standpoint. On the other hand, high power allows for sufficiently low expansion ratios. Nevertheless, considering the typical power budget of a cubesat, it is better to reduce these values as much as possible. Based on these observations, a trade-off is performed. A power of 12.5 W and an area ratio of 11 are adopted in the design, with a final temperature in the heating chamber of 440.25 K. The selected design parameters are compliant with most of the real cases seen in literature [19] [21] [26] .

Once the nominal design is frozen, the sizing can be defined. The results are reported in Table 12.

Parameter	D_{hc}	D_e	D_t	$D_{orifice}$	L_d	L_c	L_f	L_p	L_{hc}
Value [μ m]	280	383	115	109	232	82.3	714	2000	10000

Table 12: Geometrical dimensions of the VLM system components expressed in micrometers.

The obtained dimensions are consistent, in terms of lengths, with those found in the literature [21] [30]. The values found for the residence time are about 36 ms, which is reasonable for a heating chamber of this kind. As for diameters, it is more difficult to find comparable values in other studies, since axisymmetric geometries are not commonly used. However, once the areas are computed, the validity of the results becomes evident, specially when compared with values reported in literature [30]. In addition, the volumes involved are reported in Table 13.

Parameter	Value [mL]
Water Volume	2.1
Gas Volume	3.4
Tank Volume	5.5

Table 13: Volume values of water, gas, and tank capacity.

The results still remain acceptable, given the volume of a 3U cubesat and the required ΔV . The propulsion unit occupies the 0.18% of the entire available space.

The architecture of the propulsion unit is well represented by Figure 7. The adopted configuration involves a single thruster, which enables to perform the station keeping of the spacecraft. Nevertheless, by adding other two thrusters also the attitude control could be performed, provided that the thrusters are activated subsequently, otherwise the needed power would be increased. In Figure 7 the main components are highlighted. Among these, a detailed analysis is dedicated to the pressure regulating valve between the tank and the feeding lines. The aim of this analysis is to make explicit the relation between the discharge coefficient, the area of the valve and the pressure in the tank, which decreases during the blowdown. The output of this analysis is reported in Figure 9. It is evident that, as pressure in the tank decreases, the valve area must increase to keep a constant volumetric flow rate of water into the feeding lines. As this process occurs, the Reynolds number of the pipe flow increases, and the discharge coefficient increases as well.

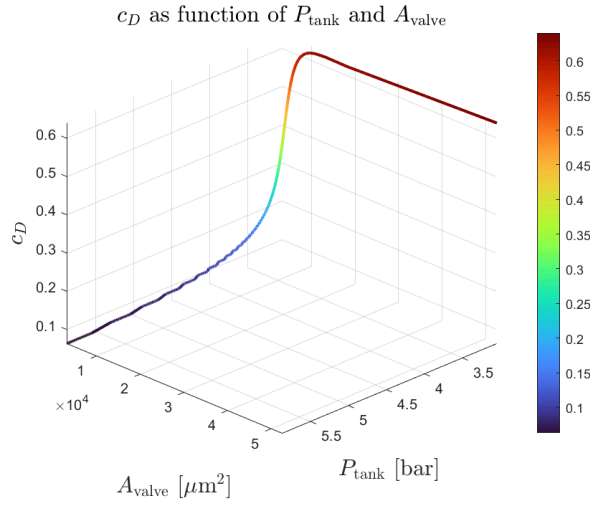


Figure 9: Discharge coefficient C_d as a function of tank pressure P_{tank} and valve cross-sectional area A_{valve} .

The main performances of the designed VLM are showcased in Table 14:

$\mathbf{F}$ [mN]	$\mathbf{c_F}$	$\mathbf{c^*}$ [m s ⁻¹]	$\mathbf{I_{sp}}$ [s]
5	1.59	675.5	109.5

Table 14: Main performance parameters for VLM.

The Knudsen number turns out to be an important parameter when considering miniaturized propulsion systems, since it represents how rarefied the flow is. $Kn < 0.01$ indicates continuous flow, slip condition arises when $0.01 < Kn \leq 0.1$, transition happens for $0.1 \leq Kn < 10$, and free molecular regimes can be observed for $Kn > 10$. As can be seen in Table 15, the values of the Knudsen at the inlet of the convergent are low, they grow little in the throat, still remaining in the continuous regime, and they grow at the outflow, consistent with the expansion of the flow.

Parameter	$\mathbf{Kn}$
Convergent	2.2×10^{-4}
Throat	8.6×10^{-4}
Divergent	6.5×10^{-3}

Table 15: Knudsen number at different nozzle sections.

3.4 Monte Carlo Analysis

The components of the VLM propulsion system are affected by uncertainties that directly depend on the technological process used for their construction. These uncertainties are connected to a variation in the overall performance parameters of the engine, in particular the thrust and the specific impulse. It is the purpose of this section to use the Monte Carlo approach to make clear how the uncertainties on the characteristic dimensions of the propulsion system impact on its performances. The Monte Carlo approach is applied on the specific impulse instead of the single impulse bit since the characteristic time for the latter would be the valve opening time, which can reasonably be considered constant. Consequently, the specific impulse would be simply a scaling of the thrust and is therefore not significant.

The dimensions used to apply the Monte Carlo analysis are the diameter of the injection orifice, $D_{orifice}$, the throat diameter, D_t , and the power generated by the heater, $\dot{Q}$. The mean values of these quantities and their standard deviations are reported in Table 16.

Parameter	Mean Value	Standard Deviation	Units
$D_{orifice}$	108.4	0.3	μm
D_t	115.4	3.5	μm
$\dot{Q}$	12.5	0.4	W

Table 16: Monte Carlo inputs for VLM.

The values relative to the standard deviations of the two diameters are recovered from [27], even though it is noteworthy to highlight that the nozzle geometry analysed in the aforementioned paper is linear, whereas the one considered for the present analysis is axisymmetric.

As for the standard deviation of the power supplied by the heater, it should be noted that this value has not been found in the literature, therefore the typical standard deviation on molybdenum resistors has been considered, which is approximately 2.9% of the nominal value and, considering the uncertainty on the supply current to be zero, a common very reasonable hypothesis in such systems, this standard deviation has an identical impact on the power. The subsequent plots show the behaviour of the cumulative mean thrust and the specific impulse, and it can be observed the plateau reached by the curves, with a standard deviation on the thrust in the order of 0.1 mN.

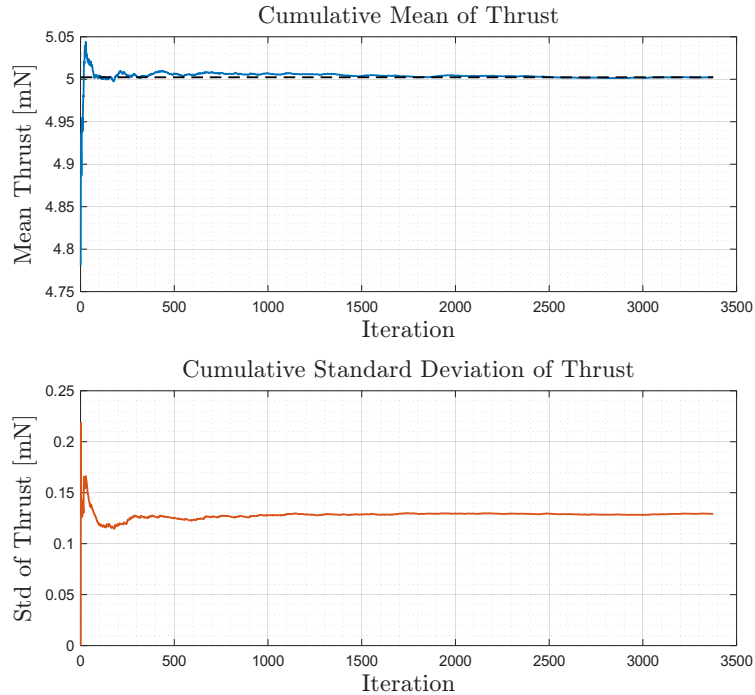


Figure 10: Thrust mean value and standard deviation for VLM.

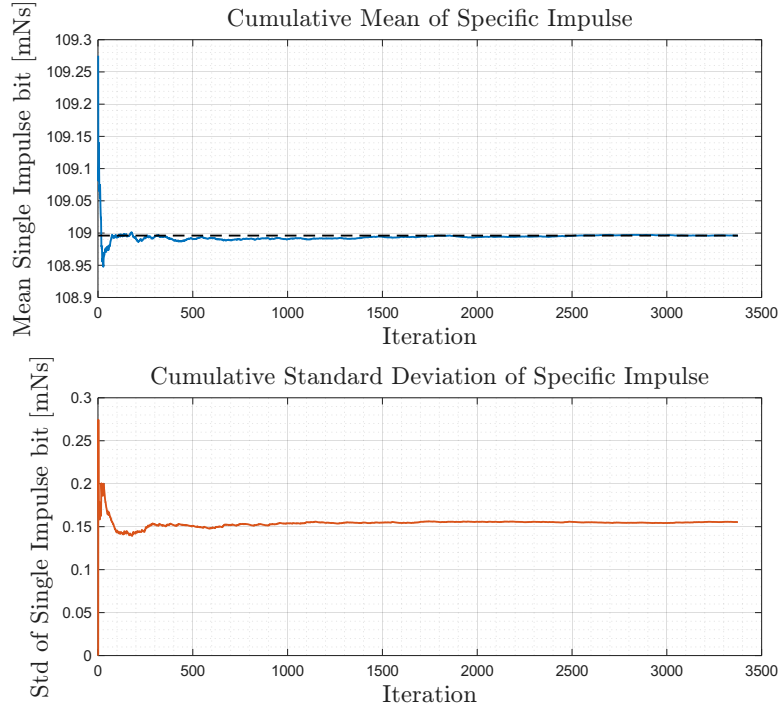


Figure 11: Specific impulse mean and standard deviation for VLM.

3.5 Future developments

- **Boiling instabilities:** boiling instabilities impact VLMs performances and limit their potential [21]. Additionally, flow oscillations can be experienced, causing unpredictability of thrust and low specific impulse efficiency. Further studies should focus on active and passive control of this phenomena in order to improve the overall performance.
- **Heating chamber:** this kind of technology requires a non-negligible amount of power in order to vaporize the water before the expansion. Therefore, future developments shall focus on a more efficient heating process, in order to lower the electric power required. This could be achieved by optimizing the performances of the heating channels, and improving the thermal insulation of the entire power unit.
- **Level of thrust:** state of the art VLM systems are capable of generating thrust in the millinewton range. Such performance may not be sufficient for all missions involving small satellites, particularly those requiring rapid attitude control or complex orbital maneuvers. Future developments will need to investigate ways to increase the available thrust, while maintaining efficiency and low energy consumption.
- **Manufacturing:** the manufacturing process is a key aspect of this kind of propulsion technology. By simplifying the production process and reducing the number of steps, VLM manufacturing could become more economically sustainable, allowing the scalability of this technology.

4 Cold gas thrusters

4.1 Introduction and literature review

Although cold gas propulsion systems were not exclusively designed for micro applications, they have proven to be a well-suited solution for small systems like cubesats, delivering great results and representing a reliable and simple approach to micro propulsion.

As pointed out in the previous brief introduction, unlike chemical propulsion, Cold Gas Micro-propulsion (CGMP) does not involve any combustion processes, relying solely on the expansion of an inert gas through a nozzle. The most popular propellant options in CGMP are Nitrogen, Hydrogen and Helium. Their properties and selection were discussed in a Delft University of Technology review [32], where Hydrogen and Helium were found to have higher specific impulses but small molecular densities, increasing the mass of the needed propellant and leading to an overall rise in the mass of the propulsion system. Such limitations can lead to the choice of a nitrogen-based propulsion system as selected by several applications [33] [34].

When analysing CGMP systems, different architecture solutions have been explored and evaluated. LISA Pathfinder CGT [33] chose a nitrogen-based system, which was particularly considered in this model for the presence of a plenum. It is an additional element, mainly used to make the flow regular and typically implemented before the gas flow is injected inside the nozzle.

The Adelis-SAMSON architecture [35] did not involve the presence of a plenum, but rather used both a high pressure regulator and a low pressure regulator valve, which is particularly interesting as it allows a fine pressure regulation without the need for an intermediate storage stage.

Another significant component is the filter, as it prevents contaminants and impurities from entering the main propulsion components, such as the nozzle. For this reason, the MEMS CGMP architecture, developed by NanoSpace was considered [36], and further in-depth analysis on micro filters was taken from other sources [37].

In addition to filtering, another aspect of great interest for CGMP is valve sealing and leakage, which can compromise the system's efficiency and its overall performance. In this context, Louwerse's thesis [38] analysed different types of valve technologies to overcome these challenges, focusing on minimizing leakage and optimizing the performance of the CGT.

As a final element of the architecture, due to the reduced dimensions of a micro-propulsion unit, the injector device was considered as a simple valve with piezoelectric actuation [38]. For the computation of such part, specifically for the pressure drop, Louwerse's thesis was not sufficient; therefore a data sheet was explored [39], keeping attention to the fact that it was intended for larger systems. It was nonetheless used to consider a worst case scenario. Having outlined all the possible solutions reviewed in the literature, the typical architecture of such system consists of a tank, a pressure regulator, a filter, a plenum, an injection valve and a nozzle.

Finally, attention was given to manufacturing technologies employed to build some of the micro components previously presented, particularly focusing on the fabrication of micro-holes such as those used in nozzle construction [40]. Micro Electro Chemical Machining (ECM) has been proven to be of great interest for this field because of its high performances on smoothness, three-dimensional micro-structures feasibility, and overall outstanding properties of the machined pieces. In the ECM process, the workpiece is either the anode or cathode and it is separated by a thin layer of flowing electrolyte from the tool electrode.

4.2 Modelling approach

After consulting different configurations, the architecture considered is shown in Figure 12, which takes into account all the main elements of a CGT and assumes the implementation of a single thruster for station-keeping reasons.

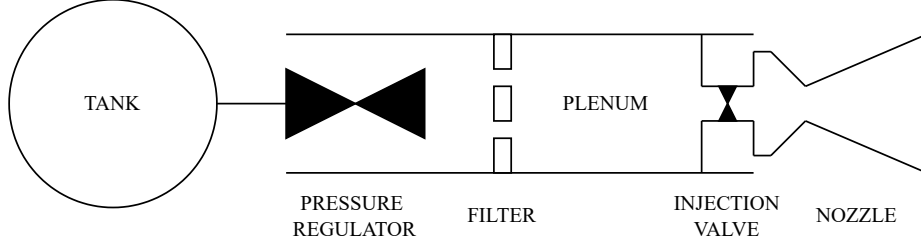


Figure 12: Cold gas thruster architecture.

The workflow adopted starts by sizing the nozzle and then moves upstream to design the other elements. However, it is necessary to assume several data to complete the configuration, and to formulate three hypotheses regarding the modelling approach for Nitrogen:

- **Continuum flow:** Section 2.2;
- **Perfect gas:** it is fair to assume Nitrogen as a perfect gas;
- **Steady-state operations:** Section 3.2.

The sizing of the nozzle requires selecting the expansion ratio, the inlet pressure and Mach number. The values used are reported in Table 17.

Property	Value
ε	10
M_{in}	0.1
P_{in}	0.18 MPa
M_{mol}	28.0134 g mol ⁻¹

Table 17: Assumed values for CGT.

The choice of the expansion ratio is reasonable considering the typical ranges for cold gas micro-thrusters [38] [41]. Also the values for P_{in} and M_{in} are consistent with simulations [42]. From ε , it is possible to compute the pressure ratio through Eq (2.2), where total pressure is P_{in} . The computation of P_e is then straightforward. After that, the thrust coefficient is derived from Eq (2.3) without considering λ , thus retrieving the ideal parameter.

Since the nominal thrust from requirements takes into account the 2D losses and the nozzle is not adapted, solving a linear system is necessary to recover the variables $\dot{m}_p$, A_t , u_e and F_{id} .

$$\begin{cases} F_{nom} = \lambda \dot{m}_p u_e + A_e P_e \\ A_e = \varepsilon A_t \\ F_{id} = \dot{m}_p u_e + A_e P_e \\ A_t = \frac{F_{id}}{P_{in} c_F} \end{cases} \quad (4.1)$$

Where λ is computed by assuming the divergent angle of the nozzle $\alpha = 20^\circ$ [43]. The model also considers the viscous losses due to the displacement thickness, thus the discharge coefficient of the nozzle is retrieved using Eq (2.3), in which as a first estimation $Re' = Re_t$, and the mass flow rate is then corrected:

$$\dot{m}_{real} = c_D \cdot \dot{m} \quad (4.2)$$

Now, the specific impulse of the propulsion system is derived:

$$I_{sp} = \frac{F_{nom}}{\dot{m}_{real} \cdot g_0} \quad (4.3)$$

From the Mach number and the perfect gas law equation, the inlet area is obtained:

$$A_{in} = \frac{\dot{m}_{real}}{u_{in} \cdot \rho_{in}} \quad (4.4)$$

Considering the convergent angle of the nozzle $\beta = 20^\circ$ [43], the nozzle can be completely sized.

The injection in the nozzle is performed by a valve which is modelled as an orifice. The component is considered adiabatic because the passage in the valve is supposed fast as the orifice represents a restriction with respect to the dimension of the system, as shown in Figure 12. In addition, the gas is assumed to be incompressible, the discharge coefficient is set to $c_{D, valve} = 0.7$ and the pressure losses across the valve are $\Delta P_{valve} = 0.15$ MPa [39].

$$P_{up, valve} = P_{in} + \Delta P_{valve} \quad (4.5)$$

$$T_{up, valve} = T_{in} \left(\frac{P_{in}}{P_{up, valve}} \right)^{\frac{1-k}{k}} \quad (4.6)$$

From the perfect gas law, the density of the gas is computed, so the area of the valve can be derived:

$$A_{valve} = \dot{m}_{real} \left(c_{D, valve} \sqrt{2 \Delta P_{valve} \rho_{up, valve}} \right)^{-1} \quad (4.7)$$

The assumption of incompressible flow is then justified a posteriori by the determination of the Mach number in the valve. Moreover, the Reynolds number is computed since it provides relevant information about the discharge coefficient in the valve. From literature reports, it is understood that when Reynolds is sufficiently high, the discharge coefficient in orifices shows an asymptotic behaviour and reaches values around $0.6 \div 0.7$ [44] [45]. In this case the Reynolds number computed guarantees the validity of the hypothesis.

Knowing the value of specific impulse, the total ΔV required and the wet mass of the satellite, the propellant mass is evaluated through the Tsiolkovsky equation. Taking into consideration the leakage of the valve, literature provided a value of $leak = 1.6 \times 10^{-4} \text{ cm}^3 \text{ min}^{-1}$ [38]. The real mass of the propellant is then derived, expanding the leaked mass for the lifetime of the mission and considering the density of the fluid before the injection.

$$m_{leak} = t_{mission} \cdot leak \cdot \rho_{plenum} \quad (4.8)$$

$$m_{prop, tot} = m_{prop} + m_{leak} \quad (4.9)$$

Following the architecture of the system, the plenum is encountered and as its main function is to regularize the gas, it is considered that the properties inside this component remain constant. For this reason, the pressure and the temperature of the plenum are equal to:

$$P_{plenum} = P_{up, valve} \quad (4.10)$$

$$T_{plenum} = T_{up, valve} \quad (4.11)$$

Given the low velocities of the gas from this point on, an isothermic approach is deduced and so temperatures will be further regarded as equal to T_{plenum} .

The characterization of the filter is made by considering its porosity $p_{filter} = 0.116$, the radius of its micro holes $r_{hole,filter} = 2.5 \mu\text{m}$ and its thickness $t_{filter} = 90 \mu\text{m}$ [37]. The area of passage of the filter is assumed to be at least 100 times larger than the nozzle throat [38] and so it is computed.

$$A_{filter} = 100 \frac{A_t}{p_{filter}} \quad (4.12)$$

For the computation of the pressure drop through the filter, Poiseuille's pipe flow formula [37] can be used along with Sutherland's law to retrieve the viscosity of the gas.

$$P_{up,filter} - P_{down,filter} = \frac{8 \dot{m}_{real} R T_{filter} t_{filter} \mu_{filter}}{P_{up,filter} p_{filter} A_{filter} r_{hole,filter}^2} \quad (4.13)$$

It is noteworthy that $P_{down,filter} = P_{plenum}$ so in Eq (4.13), the only unknown is $P_{up,filter}$.

To model the pressure difference across the pressure regulator, literature regarding typical ranges for the initial pressure of the tank was consulted, and the decided value is $P_{i,tank} = 160 \text{ bar}$ [35]. So the maximum pressure drop through the valve is estimated as:

$$\Delta P_{regulator} = P_{i,tank} - P_{up,filter} \quad (4.14)$$

To guarantee the proper functioning of the system, it is necessary to provide an appropriate pressurization during all the process. For this reason, an additional quantity of nitrogen is considered inside the tank, so that even after the propellant necessary to reach the required ΔV is expelled, the pressure inside the tank is still higher than the one after the pressure regulator. In order to achieve this, it is supposed that the final pressure in the tank shall be $P_{f,tank} = 2P_{up,filter}$. Assuming that the release of the propellant happens at constant temperature and that the volume does not change, a formula to compute the additional propellant is retrieved:

$$m_{f,tank} = \frac{M_{prop,tot}}{\frac{P_{i,tank}}{P_{f,tank}} - 1} \quad (4.15)$$

Now it is possible to calculate the total propellant mass that needs to be embarked:

$$m_{embarked} = m_{prop,tot} + m_{f,tank} \quad (4.16)$$

The sizing of the tank is performed:

$$V_{tank} = \frac{m_{embarked} R T_{plenum}}{P_{i,tank}} \quad (4.17)$$

The volume occupied by the propulsion unit is estimated considering that the length of the feeding line is $L_{feeding,line} = 5D_{filter}$, which is a reasonable assumption for a CGMP system [34], and the diameter of the whole feeding line is constant and equal to D_{filter} .

$$V_{tot} = V_{nozzle} + V_{tank} + V_{feeding,line} \quad (4.18)$$

The cold gas propulsion unit is designed to be a low cost system because of the low power required. In the analysed configuration, the only power-fed element is the injection valve, since the pressure regulator is assumed to work passively. A representative value is $W_{valve} = 350 \text{ mW}$ [38]. Nevertheless, other studies suggest that using a piezoelectric actuator to activate the valve can lead to negligible power consumption [46]. Pipe losses can be taken into account when nitrogen is flowing in the plenum. However, in this model, pipe losses are neglected, since their computation showed values in the order of 10^{-2} Pa .

4.3 Results and discussion

The results of the sizing of the nozzle are reported in Table 18.

Property	Value [μm]
D_t	149
D_{in}	366
D_e	470
L_{conv}	405
L_{div}	442

Table 18: Nozzle dimensions.

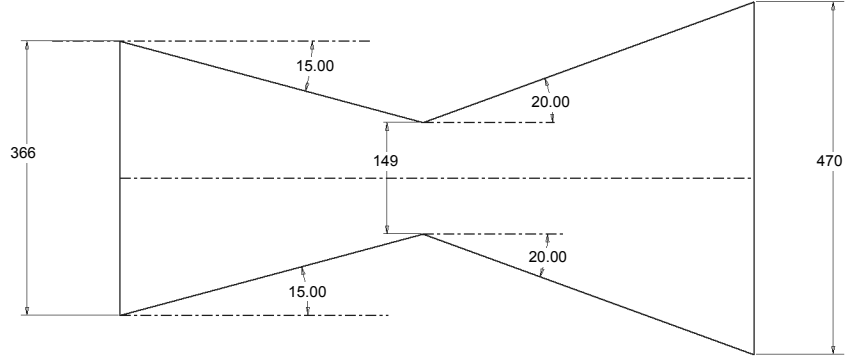


Figure 13: Nozzle geometry. [Angles in $^\circ$ & diameters in μm].

Other important parameters, including the specific impulse, the area of the feeding line, the propellant mass and the sizing of the tanks are listed in Table 19 and Table 20.

Property	Value
$D_{feed,line}$	4.4 mm
$M_{embarked}$	3.5 g
M_{prop}	3.4 g
I_{sp}	67.8 s
c_F	1.60

Table 19: Results for CGT.

Property	Value
V_{tank}	19.9 cm^3
$V_{feed,line}$	261.5 mm^3
V_{nozzle}	0.1 mm^3
V_{total}	20.1 cm^3

Table 20: Volume budget for CGT.

Before discussing the outcome of the sizing, it is important to clarify a significant continuum assumption made during the computation. Due to the small dimensions of micro-propulsion components, such hypothesis is not always guaranteed. To confirm this supposition, the Knudsen number was computed to study the rarefaction of the gas and it was observed to be 4.4×10^{-4} , which is well beyond the boundary value of 0.01. This result confirms the validity of the assumption.

As can be seen from the computation and sizing of the system, results show a significant consistency with values commonly encountered in the consulted literature. From Table 20, it is clear that the majority of the volume is occupied by the tank. Typical values for small satellites show that the solution presented and sized is suitable for 1U cubesats, thus largely satisfying the initial requirement for a larger 3U system, since the propulsion unit occupies 0.67% of the whole 3U satellite.

The specific impulse remains limited to around 70 s as it is reported in several sources [47] [48]. That implies that the system consumes more propellant for a certain amount of thrust and can be partially attributed to the absence of a heating mechanism, which would add thermal energy and increase the overall performance of the system. Despite the low value, this type of solution is particularly suitable for nanosatellites and cubesats primarily for small ΔV , as requested for the sizing.

Similarly, the nozzle geometry falls within commonly reported ranges and its sizing can be validated by literature [38]. Particular attention can be given to the uncertainties of the profile of such component. For micro-nozzles, it is difficult to manufacture perfectly smooth walls relative to the small dimension of the system. Such roughness can lead to an increase in friction and boundary layer thickness. From literature [43], it has been discussed how particular geometries of the surface roughness affect differently the performance of the components because of the alternating increase and decrease in the nozzle area, causing the formation of weak shocks that reduce the velocity of the gas and cause a loss of total pressure.

For this reason, the next section will examine this aspect to further validate the sizing of the propulsion system.

4.4 Monte Carlo Analysis

The Monte Carlo Analysis is performed to assess the effects on the performance of the system caused by the uncertainties on the diameter of the throat and of the injection valve. The standard deviation values are obtained from literature focused on micro holes machining [40]. The values used are listed in Table 21.

Parameter	Mean value	Standard deviation	Units
D_{valve}	232	3	μm
D_t	149	4	μm

Table 21: Monte Carlo inputs for CGT.

The nozzle geometry, particularly the exit diameter D_e and the inlet diameter D_i are also taken as an input in the simulation without uncertainties, since the geometry of the nozzle is thought to be fixed. In a first approximation, the discharge coefficient of the nozzle c_D , the temperature and the pressure after the plenum are also deemed as constant. To recover the performance parameters influenced by the uncertainties, it is necessary to solve a system imposing the conservation of mass, the formula for the expansion ratio and the adiabatic transformation in the valve.

The performance parameters analysed are the thrust and the specific impulse, which are found to be the most significant. Another meaningful parameter is the single impulse bit, that can be computed as $SIB = F \cdot t_{opening, valve}$. Since it is acceptable to consider $t_{opening, valve}$ constant in a first approximation, this value would result in a scaling of the thrust behaviour and it would not provide any further information about the performances.

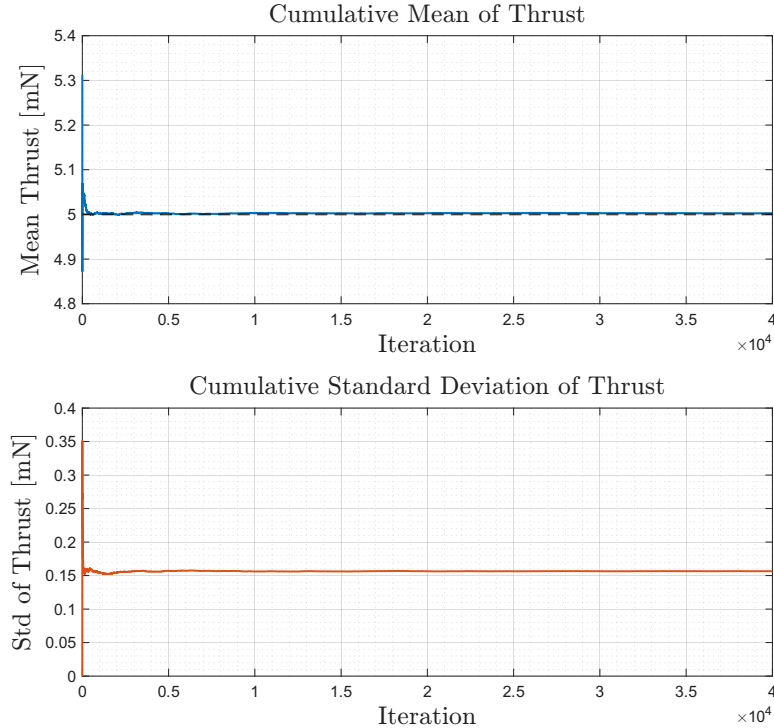


Figure 14: Thrust mean value and standard deviation for CGMP.

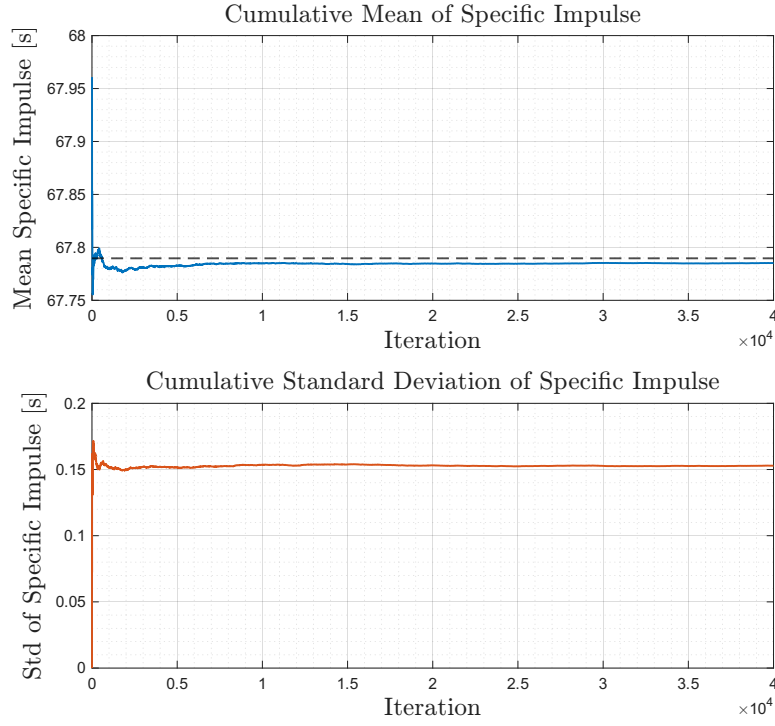


Figure 15: Specific impulse mean value and standard deviation for CGMP.

The results of the simulation are quite satisfactory as both parameters showed convergence to the nominal value with small (at least one order of magnitude less) standard deviations.

4.5 Future developments

- **Valve leakage:** during the research of this CGMP system, a special emphasis was put on the leakage of the valves, as previous research proves it to be a significant problem in this micro propulsion system [38]. A possible answer can be provided, relying on a revolutionary sealing mechanism of the valve using a flexible membrane and specifically shaped valve seats. However, the issue seems to be one of the main open challenges of CGMP.
- **Cold Gas Generators (CGG):** in a conventional system, most of the mass of the propulsion unit is consumed by the tank, which needs a thick wall to withstand the high pressure. A possible solution is the application of CGG, which allow for a low storage pressure by storing solid material, that releases the gas inside the system when ignited [49].
- **Low pressure storage:** in the proposed configuration, the propellant was stored at high pressure. However, a fair alternative can be using a low pressure storage. This solution would decrease the overall mass of the propulsion system since the presence of a pressure regulator can be neglected and the thickness of the wall tank can be further reduced.

5 Comparison and conclusion

In this final section, the different propulsion systems designs are compared. Main performance, volume and energy budget parameters are grouped in Table 22.

Parameter	MSPT	VLM	CGT
Volume [%]	0.38	0.18	0.67
Power [W]	N/A	12	0.350×10^{-3}
I_{sp} [s]	223.3	109.5	67.8
m_P [g]	1.2	2.1	3.4
c_F [-]	1.64	1.59	1.60

Table 22: Parameter comparison.

All these three possible solutions are modelled to be mounted on a 3U CubeSat, therefore the volume occupied by the propulsion unit is of main importance. In Table 22, the encumbrance percentages with respect to the volume of the whole satellite ($10 \times 10 \times 30$ cm) are reported and the VLM turns out to be the most compact configuration.

However, VLM requires a considerable amount of power. Although no power budget has been computed for MSPT technology, typical values to ignite the propellant are around 2.6 W [1]. This value is referred to a single thruster ignition, thus also the number of simultaneous ignition must be kept under observation. Consequently, MSPT and CGT become competitive technologies since they require a lower level of power.

Other selection criteria could be based on propulsive parameters, like the specific impulse. Even when dealing with micro-thrusters, solid propulsion retrieves the highest value of I_{sp} , consequently less mass shall be embarked. This could be a significant benefit for small systems, like cubesats, in which the embarked mass is strongly related to the overall cost of the mission. Furthermore, as it is shown in Table 22, the thrust coefficient turned out to be very alike for all the three modelled solutions. Therefore, despite the different designs, the nozzle seems to perform in similar ways.

From a technological viewpoint, it needs to be highlighted that the VLM and CGT are the most complex configurations since they require miniaturised valves. They represent still today a challenge and are more prone to failure. As a consequence, the MSPT solution is preferred due to the relative simplicity and higher reliability. On the other hand, this solution introduces mechanical moving parts, which allow for more flexibility but less reliability, and could also lead to an increase of costs.

In conclusion, this report provides the applications and the limitations of the mathematical models normally used for the preliminary design of conventional space propulsion. Additionally, the performances of each solution are shown and compared with real experimental results, enabling a better overview of the phenomena associated to micro-scales. The small discrepancies between real experimental data and those obtained by the adopted models suggest that the latter are still capable of predicting the functioning of micropropulsion systems, even when they are applied at the limit of their validity.

Appendix A: Authorship Declaration

The list reported below is approximate. The team wants to emphasize that everyone gave their own contribution to the project and even though there was an initial distribution of tasks, everyone helped on everything. Overall, everybody took part in the brainstorming and briefing process of the project.

Esposito Ginevra : literature review on CGMPs, parameters research for the modelling and comparison, accuracy values investigation for Monte Carlo analysis.

Fossati Francesco : literature review, modelling and performance analysis of Vaporizing Liquid Micro-Thrusters.

Fulgheri Matteo : literature review on CGMPs, modelling and Monte Carlo simulation of CGMPs.

Ghilardini Sara : literature review on CGMPs, modelling and Monte Carlo simulation of CGMPs.

Gjoni Endi : literature review, modelling and performance analysis of Vaporizing Liquid Micro-Thrusters.

Jayakody Arachchilage Senura Lisan Martino : literature review on MSPTs, modelling of Solid propulsion system and related Monte Carlo Analysis.

Mauroner Niccolò : literature review, modelling analysis of Vaporizing Liquid Micro-Thrusters and Monte Carlo analysis.

Navarro Zea Mario : literature review on MSPTs, modelling of Solid propulsion system and related Monte Carlo Analysis.

Usage of LLMs: for the generation of the logo "Thrustbusters" and BibLaTeX references codes.

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